

rapport

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1 Customer Segmentation and Behavioral Analysis Using Unsupervised Learning

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Program: Data Science – Master Level

Project Type: Exploratory Data Analysis & Customer Segmentation

Dataset: Marketing Campaign Dataset

1.1 1. Introduction

Understanding customer behavior is essential for businesses aiming to improve marketing performance and customer retention. Customer segmentation helps divide a customer base into homogeneous groups, allowing companies to personalize strategies, optimize marketing resources, and increase conversion rates.

In this project, we perform customer segmentation using **unsupervised machine learning** techniques. Since the dataset does not include labeled customer groups, clustering methods are applied to uncover hidden patterns based on demographic and behavioral features.

The objectives of this study are to: - Explore customer characteristics and behaviors - Identify meaningful customer segments - Interpret segments from a business perspective - Propose data-driven marketing strategies

1.2 2. Dataset Description

The dataset contains information about: - **Demographics:** age, income, education, marital status, family size - **Purchasing behavior:** expenses, number of purchases, product categories - **Marketing interaction:** campaign acceptance and response - **Engagement metrics:** recency, time as a client, complaints

The dataset initially contained 2240 customers and 29 features.

1.3 3. Data Cleaning and Preparation

The following preprocessing steps were applied:

- Removed missing values in the **Income** column (1.1% of the data)
 - Dropped non-informative features with no variance (**Z_CostContact**, **Z_Revenue**)
 - Optimized data types to reduce memory usage
 - Converted date columns to datetime format
 - Checked and confirmed no duplicate rows
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1.4 4. Feature Engineering

To improve analysis quality, new features were created:

- **Age**: calculated from birth year
- **Days_is_client**: number of days since customer enrollment
- **Kids**: total number of children and teenagers
- **Expenses**: total spending across all product categories
- **TotalNumPurchases**: total number of purchases across all channels
- **TotalAcceptedCmp**: number of accepted marketing campaigns

Categorical features were standardized: - Marital status → *Single* / *Partner* - Education → *Undergraduate* / *Graduate* / *Postgraduate*

1.5 5. Exploratory Data Analysis (EDA)

1.5.1 Univariate Analysis

- Most customers are graduates, partnered, and have 0–1 child
- Expenses are right-skewed: few customers account for high spending
- Most customers have never accepted marketing campaigns
- Complaints are rare

1.5.2 Bivariate & Multivariate Analysis

Key relationships observed: - Strong correlation between **income**, **expenses**, and **purchases**
- Higher income leads to higher spending and purchase frequency - Education and family size influence behavior mainly through income - Age and tenure alone do not explain customer value - Marital status has a weak effect on spending behavior

1.6 6. Feature Encoding and Scaling

- One-hot encoding was applied to categorical variables
 - Features were scaled using **StandardScaler** since PCA and K-Means are distance-based methods
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1.7 7. Dimensionality Reduction

1.7.1 Principal Component Analysis (PCA)

PCA was applied to reduce dimensionality, remove multicollinearity, and speed up clustering.

From the scree plot, **10 principal components** were retained, explaining **80% of the total variance**, which is sufficient for clustering and visualization.

1.7.2 t-SNE Visualization

t-SNE was used to visualize the data in 2D: - Before clustering: clusters were not clearly visible - After clustering: clear and distinct clusters appeared, confirming effective segmentation

1.8 8. Customer Segmentation (K-Means)

The **Elbow Method** was used to determine the optimal number of clusters.

The elbow point was observed at **k = 3**, indicating three meaningful customer segments.

K-Means clustering was applied on the PCA-reduced data, resulting in well-separated clusters with minimal overlap.

1.9 9. Cluster Profiling and Interpretation

1.9.1 Cluster 0 – High-Value Loyal Customers (VIPs)

- Age: 45–60
- Income: highest
- Family: mostly partnered, 0–1 kids
- Behavior: frequent purchases, high spending
- Business value: highest lifetime value

1.9.2 Cluster 1 – Mid-Value Growth Customers

- Age: 40–55
- Income: medium
- Family: mixed, 1–2 kids
- Behavior: moderate spending and engagement
- Business value: strong growth potential

1.9.3 Cluster 2 – Low-Value / Price-Sensitive Customers

- Age: 30–45
 - Income: lowest
 - Family: mostly single or small families
 - Behavior: infrequent purchases, low spending
 - Business value: high churn risk
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1.10 10. Business Strategies and Recommendations

1.10.1 Cluster 0 – VIP Customers

- Launch loyalty programs and exclusive rewards
- Offer premium bundles and upsell products
- Personalized communication
- Early access to new products
- Strong retention focus

1.10.2 Cluster 1 – Growth Customers

- Targeted promotions and discounts
- Bundle offers and cross-selling
- Educational campaigns
- Personalized recommendations
- Incentives for repeat purchases

1.10.3 Cluster 2 – Low-Value Customers

- Entry-level products and starter packages
 - Automated low-cost campaigns
 - Onboarding programs
 - Limited-time discounts
 - Cost-efficient retention strategies
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1.11 11. Conclusion

This project successfully identified **three distinct customer segments** using unsupervised learning techniques. The analysis showed that customer value is mainly driven by **income, spending, and purchasing behavior**, while demographic factors such as age and marital status have a secondary influence.

By combining EDA, PCA, t-SNE, and K-Means clustering, we obtained meaningful and actionable customer segments that can directly support marketing decision-making. The results demonstrate the value of data-driven segmentation for improving marketing efficiency and customer relationship management.
